InsureFlow – Insurance Management Platform Documentation

Introduction

InsureFlow is a full-stack Insurance Management System developed to simplify insurance policy management through a secure and user-friendly digital platform. The application enables customers to purchase insurance policies, make premium payments, submit insurance claims, and monitor their insurance activities through a centralized dashboard. Administrators can manage insurance plans, review customer claims, and oversee the overall system.

The project demonstrates enterprise-level backend development using **Java, Spring Boot, Spring Security, JWT Authentication, PostgreSQL**, and a modern **React** frontend. The application follows a layered architecture that emphasizes scalability, maintainability, and security while implementing industry-standard RESTful APIs.

Design Inspiration

This project is inspired by modern insurance management platforms such as:

1. PolicyBazaar

- Online insurance management
- Digital policy purchase
- Premium comparison
- Customer dashboard

2. ACKO Insurance

- Digital-first insurance experience
- Simplified claim process
- Online premium payments
- Customer-centric workflow

3. InsureFlow Custom Implementation

- JWT Authentication
- Policy Management
- Premium Payment System
- Insurance Claim Management
- Role-Based Authorization
- PostgreSQL Database
- React Dashboard
- Spring Boot REST APIs

Roles

Customer

- Register account
- Login securely
- Browse insurance plans
- Purchase insurance policies
- View purchased policies
- Pay insurance premiums
- Submit insurance claims
- Track claim status
- View payment history

Administrator

- Login securely
- Create insurance plans
- Update insurance plans
- Delete insurance plans
- View all purchased policies
- View all premium payments
- View all claims
- Approve or reject insurance claims
- Manage insurance products

Agent *(Future Enhancement)*

- Login securely
- View assigned customers
- Review claims
- Assist customers
-

Use Cases of the Project

Key Use Cases

User Registration

- Customer creates an account
- Password encrypted using BCrypt
- User stored securely in PostgreSQL

User Authentication

- Customer logs in
- JWT token generated
- Protected APIs accessed using Authorization Bearer Token

Premium Payment

- Select purchased policy
- Make premium payment
- Store payment history
- Generate transaction details

Claim Management

- Submit insurance claims
- Enter claim amount
- Provide reason and description
- Track claim status

Administration

- Manage insurance plans
- Review customer claims
- Approve claims
- Reject claims
- View complete system information

Tech Stack

Backend

Spring Boot

- REST API development
- Dependency Injection
- Layered Architecture
- Business Logic Implementation

Spring Security

- Endpoint protection
- Authentication management
- Authorization handling
- Secure API access

JWT (JSON Web Token)

- Stateless authentication
- Secure API communication
- Authorization header validation
- Token-based access control

Spring Data JPA (Hibernate)

- ORM implementation
- Repository abstraction
- Entity management
- Database operations

PostgreSQL

- Users
- Policy Types
- Policies
- Premium Payments
- Claims

BCrypt Password Encoder

- Password Hashing
- Secure Password Storage

Frontend

React

- Component-based architecture
- Dynamic UI rendering
- API integration
- State management

Tailwind CSS

- Utility-first styling
- Responsive layouts
- Modern dashboard interface
- Faster UI development

Axios

- REST API communication
- JWT header injection
- Request interceptors

Authentication

Authentication is implemented using JWT (JSON Web Token).

Flow

User Login

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Credential Validation

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JWT Token Generated

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Authorization: Bearer <TOKEN>

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Protected API Access

Security Layers

JwtFilter

- Validates JWT tokens
- Extracts authenticated users
- Secures protected endpoints

Spring Security

- Authorization management
- Endpoint security
- Authentication context

BCrypt Password Encoder

- Password hashing
- Secure credential storage

Flowcharts

Backend Flow

Client
↓
JWT Authentication
↓
Controller Layer
↓
Service Layer
↓
Repository Layer
↓
PostgreSQL Database
↓
JSON Response

Authentication Flow

Client

↓

Register/Login Request

↓

Credential Validation

↓

JWT Token Generation

↓

Token Returned

↓

Authorization Header

↓

Protected API Access

Policy Purchase Flow

Customer

↓

Browse Policy Types

↓

Select Policy

↓

Purchase Policy

↓

Database Storage

↓

Policy Created

Premium Payment Flow

Customer

↓

Select Purchased Policy

↓

Choose Payment Method

↓

Payment Processing

↓

Payment Stored

↓

Payment History Updated

Claim Flow

Customer

↓

Select Policy

↓

Enter Claim Details

↓

Submit Claim

↓

Database Storage

↓

Status = Pending

↓

Admin Review

↓

Approve / Reject

System Architecture

React Frontend

↓

REST API

↓

Spring Security

↓

JWT Authentication

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Controller Layer

↓

Service Layer

↓

Repository Layer

↓

PostgreSQL Database

Database Design (ER Concept)

User

|

├── purchases many Policies

|

├── makes many Premium Payments

|

└── submits many Claims

PolicyType

|

└── belongs to many Policies

Policy

|

├── belongs to one User

|

├── belongs to one PolicyType

|

└── has many Premium Payments

Claim

|

└── belongs to one Policy

PremiumPayment

|

└── belongs to one Policy

Frontend Integration Flow

React Dashboard

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REST API Calls

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Spring Boot Backend

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JWT Authentication

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PostgreSQL Database

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JSON Response

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Dashboard Update

Industry Value of the Project

This project demonstrates real-world enterprise software development concepts.

Industry Applications

- Insurance Companies
- Policy Management Systems
- Digital Insurance Platforms
- Banking Insurance Products
- Financial Service Providers
- Enterprise Insurance Portals

Key Concepts Implemented

- JWT Authentication
- Spring Security
- RESTful API Development
- Layered Architecture
- DTO-Based Design
- CRUD Operations
- Role-Based Authorization
- Global Exception Handling
- Validation
- PostgreSQL Database Design
- Full-Stack Development

Technologies Used

Technologies Used

Backend

- Java 21
- Spring Boot
- Spring Security
- JWT
- Spring Data JPA
- Hibernate
- PostgreSQL
- Maven
- Lombok
- BCrypt Password Encoder

Frontend

- React
- Tailwind CSS
- Fetch API / Axios
- Vite

Integration Overview

React Frontend

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REST API

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Spring Security

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JWT Authentication

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Controller

↓

Service

↓

Repository

↓

PostgreSQL Database

↓

JSON Response

Conclusion

InsureFlow successfully demonstrates the design and implementation of a secure, scalable, and enterprise-ready Insurance Management System.

Key Achievements

- Secure JWT Authentication
- Role-Based Authorization
- Insurance Policy Management
- Premium Payment System
- Insurance Claim Management
- PostgreSQL Relational Database
- Global Exception Handling
- Request Validation
- RESTful API Development
- Layered Software Architecture
- React Dashboard Integration

The project draws inspiration from modern digital insurance platforms such as **PolicyBazaar** and **ACKO Insurance**, while introducing a custom implementation focused on secure authentication, policy lifecycle management, premium payment processing, and claim handling.

InsureFlow showcases industry-standard backend development practices using **Spring Boot, Spring Security, JWT Authentication, PostgreSQL, React, REST APIs, DTO-based architecture, and role-based access control**, making it a practical, scalable, and portfolio-worthy full-stack application.